

Problem

A computer manufacturer has to plan the next season production of 3 types of computers. Expected benefits for each model unit are $b_i, i = 1, \dots, 3$ currency units. The company will buy now all the components for building the computers, excluding the memory modules and hard drives, which will be purchased in the future at the time of building the computers, since it is expected their prices will be lower. Each type of computer requires a number of memory modules and hard drives: 2 memory modules and 1 hard drive for the first type; 2 memory modules and 2 hard drives for the second type; and 1 memory modules and 2 hard drives for the third type. However, the available number of memory modules and hard drives in the market at the time of building the computers will be independent random variables ξ_1 and ξ_2 . For each required, but not available, memory module or hard drive the company estimates costs (benefit losses) of c_1 (for memory modules) and c_2 (for hard drives) currency units.

- (a) Formulate the stochastic optimization problem to be solved by the company to get the optimal production planning (number of computers per type, to maximize benefits and to reduce the costs by lack of memory modules and hard drives).
- (b) Formulate a new problem by replacing the second stage by the following chance constraint: “the probability that the required number of memory modules and hard drives is less or equal than ξ_1 and ξ_2 , respectively, has to be greater or equal than α ”. Take into account that ξ_1 and ξ_2 are independent. Simplify the chance constraint to obtain a linear expression, assuming that variables $\xi_1 - 100$ and $\xi_2 - 200$ have an exponential distribution of parameters λ_1 and λ_2 , that is, the probability distributions of ξ_1 and ξ_2 are $F_{\xi_1}(x) = 1 - e^{-\lambda_1(x-100)}$ and $F_{\xi_2}(x) = 1 - e^{-\lambda_2(x-200)}$.

Solution

- (a)
 - First stage. $x_i, i = 1, \dots, 3$: number of computers per type.
 - Second stage. $y_j, j = 1, \dots, 2$: number of required but not available memory modules and hard drives.

$$\begin{aligned} \max \quad & \sum_{i=1}^3 b_i x_i + E[Q(x, \xi)] \\ \text{s.to} \quad & x_i \geq 0, x_i \in \mathbb{Z}, i = 1, \dots, 3. \end{aligned}$$

where

$$Q(x, \xi) = \begin{aligned} \max \quad & -c_1 y_1 - c_2 y_2 \\ \text{s.to} \quad & 2x_1 + 2x_2 + x_3 - y_1 \leq \xi_1 \\ & x_1 + 2x_2 + 2x_3 - y_2 \leq \xi_2 \\ & (y_1, y_2) \geq 0 \end{aligned}$$

- (b)

$$\begin{aligned} \max \quad & \sum_{i=1}^3 b_i x_i \\ \text{s.to} \quad & P(2x_1 + 2x_2 + x_3 \leq \xi_1) P(x_1 + 2x_2 + 2x_3 \leq \xi_2) \geq \alpha \\ & x_i \geq 0, x_i \in \mathbb{Z}, i = 1, \dots, 3. \end{aligned}$$

Simplifying the probabilistic constraint:

$$\begin{aligned} P(\xi_1 \geq 2x_1 + 2x_2 + x_3) &= 1 - F_{\xi_1}(2x_1 + 2x_2 + x_3) = e^{-\lambda_1(2x_1 + 2x_2 + x_3 - 100)} \\ P(\xi_2 \geq x_1 + 2x_2 + 2x_3) &= 1 - F_{\xi_2}(x_1 + 2x_2 + 2x_3) = e^{-\lambda_2(x_1 + 2x_2 + 2x_3 - 200)} \end{aligned}$$

thus the chance constraint can be written as the linear constraint:

$$\begin{aligned} e^{-\lambda_1(2x_1 + 2x_2 + x_3 - 100)} e^{-\lambda_2(x_1 + 2x_2 + 2x_3 - 200)} &\geq \alpha \\ -\lambda_1(2x_1 + 2x_2 + x_3 - 100) - \lambda_2(x_1 + 2x_2 + 2x_3 - 200) &\geq \ln \alpha \\ (2\lambda_1 + \lambda_2)x_1 + (2\lambda_1 + 2\lambda_2)x_2 + (\lambda_1 + 2\lambda_2)x_3 &\leq -\ln \alpha + 100\lambda_1 + 200\lambda_2. \end{aligned}$$